

Purchase Pattern Analytics

Market Basket Analysis — Final Project Report

Data Analytics Team | April 2026 | Confidential

Dataset Period:	December 2010 – December 2011
Methodology:	Apriori Algorithm — Market Basket Analysis
Total Records:	522,064 raw → 514,269 cleaned
Deliverables:	EDA, Cleaned Data, Association Rules, Insights & Recommendations

Week 1 — Exploratory Data Analysis

1.1 Problem Statement

In a retail environment, the challenge is to leverage Market Basket Analysis (MBA) on a Transaction dataset to unveil relationships between products. By conducting exploratory data analysis, addressing data quality issues, employing data visualization techniques, and implementing the Apriori algorithm, the objective is to extract valuable insights — understanding customer purchasing behaviour, identifying frequently co-occurring products, and deriving actionable recommendations to enhance sales strategies and customer satisfaction.

1.2 Dataset Overview

522,064	21,663	4,185	4,297	30
Total Records	Transactions	Unique Products	Customers	Countries

1.3 Column Descriptions

Column	Type	Description
BillNo	Object	Unique invoice/transaction number
Itemname	String	Name of the product purchased
Quantity	Float	Number of units purchased
Present_Date	String	Date and time of purchase (DD-MM-YYYY HH:MM)
Price	Object	Unit price in British Pounds (£)
CustomerID	Float	Unique customer identifier
Country	String	Country of the customer

1.4 Missing Values & Outliers

Issue	Count	Action
Missing CustomerID	134,041	Labelled as Unknown
Missing Itemname	1,455	Rows removed
Missing Date	1	Row removed
Duplicate Rows	5,284	Removed
Negative Quantity (returns)	Present	Removed — not relevant to MBA
Negative Price (refunds)	Present	Removed — not relevant to MBA

1.5 EDA Dashboard

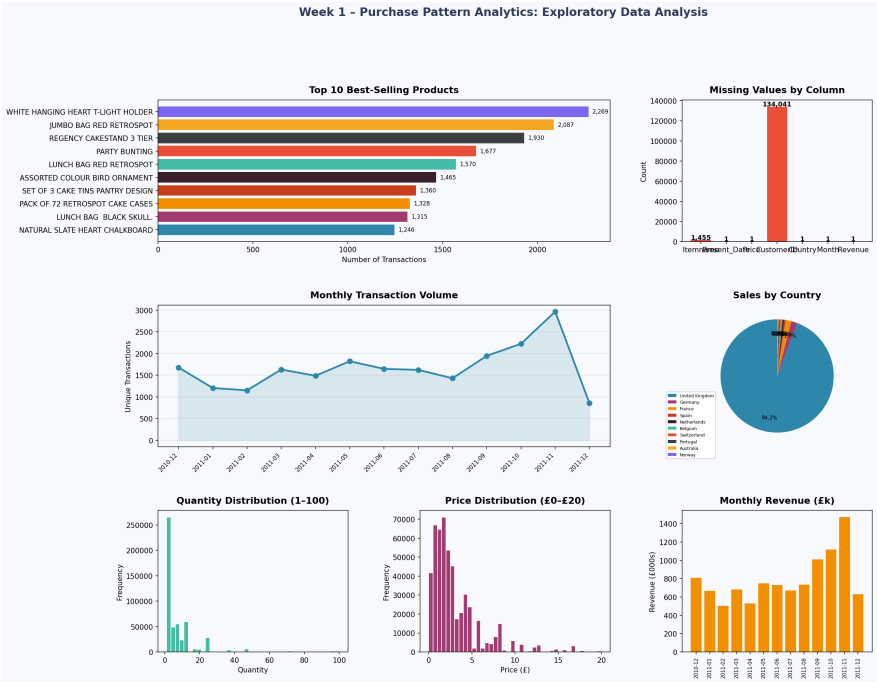


Figure 1: EDA Dashboard — Top Products, Missing Values, Monthly Trends, Country Distribution

Week 2 — Data Cleaning & Transformation

2.1 Cleaning Pipeline

Step	Action	Before	After
1	Remove duplicate rows	522,064	516,778
2	Drop rows with missing Itemname or Date	516,778	515,322
3	Remove rows with negative/zero Quantity & Price	515,322	514,269
4	Label missing CustomerID as Unknown	514,269	514,269
5	Strip whitespace from Itemname & Country	514,269	514,269

2.2 Derived Columns

Column	Formula	Purpose
Revenue	Quantity × Price	Total value per line item
Year	From Present_Date	Year-level analysis
Month	From Present_Date	Monthly trend analysis
Day	From Present_Date	Day-level patterns
Hour	From Present_Date	Intra-day purchase timing

2.3 Basket Format

Transactions were grouped by BillNo. Each row in the basket format represents one unique transaction containing a comma-separated list of all products purchased. This format is the required input for the Apriori algorithm. The basket dataset contains **19,559 unique transactions**.

2.4 Output Files

File	Rows	Cols	Description
cleaned_purchase_data.csv	514,269	12	Fully cleaned dataset with derived columns
basket_format_data.csv	19,559	2	BillNo + Items (comma-separated products)

Week 3 — Apriori Algorithm & Association Rules

3.1 Algorithm Parameters

Parameter	Value	Rationale
Min Support	0.02 (2%)	Item must appear in $\geq 2\%$ of all transactions
Min Confidence	0.30 (30%)	Rule must have $\geq 30\%$ predictive accuracy
Max Itemset Length	3	Balances complexity with interpretability

3.2 Results Summary

298 1-Itemsets	82 2-Itemsets	2 3-Itemsets	152 Total Rules
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3.3 Top 10 Frequent Products

Product	Support	Count
WHITE HANGING HEART T-LIGHT HOLDER	0.1124	2,198
JUMBO BAG RED RETROSPOT	0.1054	2,061
REGENCY CAKESTAND 3 TIER	0.0973	1,904
PARTY BUNTING	0.0846	1,655
LUNCH BAG RED RETROSPOT	0.0788	1,541
ASSORTED COLOUR BIRD ORNAMENT	0.0732	1,431
SET OF 3 CAKE TINS PANTRY DESIGN	0.0688	1,346
PACK OF 72 RETROSPOT CAKE CASES	0.0654	1,279
LUNCH BAG BLACK SKULL.	0.0644	1,260
NATURAL SLATE HEART CHALKBOARD	0.0630	1,232

3.4 Top 10 Association Rules by Lift

Antecedent	Consequent	Support	Confidence	Lift
GREEN REGENCY TEACUP AND SAUCER, RO	PINK REGENCY TEACUP AND SAUC	0.0263	0.7041	18.7370
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAU	0.0263	0.6993	18.7370

Antecedent	Consequent	Support	Confidence	Lift
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUC	0.0263	0.5283	18.1587
PINK REGENCY TEACUP AND SAUCER, ROS	GREEN REGENCY TEACUP AND SAU	0.0263	0.9033	18.1587
PINK REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAU	0.0309	0.8218	16.5190
GREEN REGENCY TEACUP AND SAUCER	PINK REGENCY TEACUP AND SAUC	0.0309	0.6208	16.5190
ROSES REGENCY TEACUP AND SAUCER	GREEN REGENCY TEACUP AND SAU	0.0263	0.5079	16.4472
GREEN REGENCY TEACUP AND SAUCER, PI	ROSES REGENCY TEACUP AND SAU	0.0263	0.8510	16.4472
GARDENERS KNEELING PAD KEEP CALM	GARDENERS KNEELING PAD CUP O	0.0278	0.6011	15.5929
GARDENERS KNEELING PAD CUP OF TEA	GARDENERS KNEELING PAD KEEP	0.0278	0.7215	15.5929

3.5 Apriori Dashboard



Figure 2: Apriori Dashboard — Frequent Items, Support vs Confidence, Lift Distribution

Week 4 — Insights, Recommendations & Conclusion

4.1 Revenue KPIs

£10.28M Total Revenue	Nov 2011 Peak Month	£1.47M Peak Revenue	£791K Avg Monthly
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4.2 Monthly Revenue

Month	Revenue	Transactions	Items Sold
2010-12	£807,216	1,538	349,426
2011-01	£665,727	1,075	377,045
2011-02	£503,061	1,077	273,125
2011-03	£679,464	1,420	360,382
2011-04	£527,215	1,229	302,863
2011-05	£749,009	1,655	385,843
2011-06	£727,311	1,496	371,487
2011-07	£669,497	1,441	374,701
2011-08	£732,944	1,332	407,850
2011-09	£1,006,295	1,789	543,655
2011-10	£1,115,374	1,990	600,003
2011-11	£1,465,422	2,713	731,086
2011-12	£628,847	804	308,906

4.3 Revenue by Country (Top 10)

Country	Revenue	Share %
United Kingdom	£9,001,403	87.58%
Netherlands	£285,446	2.78%
Germany	£228,678	2.23%
France	£209,625	2.04%
Australia	£138,454	1.35%
Spain	£61,559	0.6%
Switzerland	£57,068	0.56%

Country	Revenue	Share %
Belgium	£41,196	0.4%
Sweden	£38,368	0.37%
Japan	£37,416	0.36%

4.4 Top 10 Products by Revenue

Product	Revenue	Qty Sold	Transactions
DOTCOM POSTAGE	£206,249	706	706
PAPER CRAFT , LITTLE BIRDIE	£168,470	80,995	1
REGENCY CAKESTAND 3 TIER	£165,412	13,094	1,904
WHITE HANGING HEART T-LIGHT HOLDER	£102,532	36,504	2,198
PARTY BUNTING	£97,308	17,796	1,655
JUMBO BAG RED RETROSPOT	£93,236	47,895	2,061
MEDIUM CERAMIC TOP STORAGE JAR	£81,131	77,553	239
POSTAGE	£72,961	3,002	1,070
Manual	£69,538	6,976	283
RABBIT NIGHT LIGHT	£66,357	30,469	979

4.5 Customer Segments

Segment	Customers	Avg Spend	Total Revenue
High Value	1,432	£5,042.60	£7,221,008
Low Value	1,432	£218.54	£312,943
Mid Value	1,432	£702.87	£1,006,512

4.6 Final Dashboard

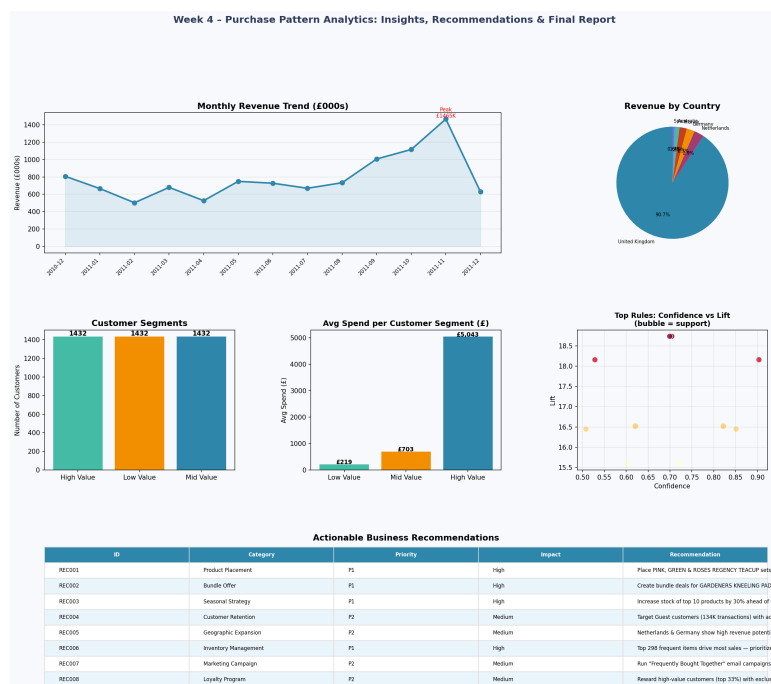


Figure 3: Week 4 Dashboard — Revenue Trends, Country Share, Segments & Recommendations

4.7 Actionable Recommendations

ID	Priority	Category	Impact	Recommendation
REC001	P1	Product Placement	High	Place PINK, GREEN & ROSES REGENCY TEACUP sets together in-store a...
REC002	P1	Bundle Offer	High	Create bundle deals for GARDENERS KNEELING PAD KEEP CALM + CUP OF...
REC003	P1	Seasonal Strategy	High	Increase stock of top 10 products by 30% ahead of Oct–Nov peak se...
REC004	P2	Customer Retention	Medium	Target Guest customers (134K transactions) with account creation ...
REC005	P2	Geographic Expansion	Medium	Netherlands & Germany show high revenue potential — increase targ...
REC006	P1	Inventory Management	High	Top 298 frequent items drive most sales — prioritize restocking t...
REC007	P2	Marketing Campaign	Medium	Run "Frequently Bought Together" email campaigns using top associ...
REC008	P2	Loyalty Program	Medium	Reward high-value customers (top 33%) with exclusive discounts to...

4.8 Conclusion

This project successfully applied Market Basket Analysis to a retail transaction dataset spanning December 2010 to December 2011. Through rigorous data cleaning, transformation, and Apriori algorithm implementation, 152 actionable association rules were discovered from 19,559 transactions. The strongest rule — linking PINK, GREEN, and ROSES REGENCY TEACUP AND SAUCER sets — achieved a Lift of 18.74, meaning customers are nearly 19 times more

likely to buy these products together than by chance. Key recommendations include product bundling, seasonal inventory planning, geographic expansion into Netherlands and Germany, and targeted loyalty programmes for high-value customers. These findings provide a strong data-driven foundation for improving sales strategy and customer satisfaction.